Linear Algebra and Calculus Lectures 1 and 2
Functions and Limits

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Functions

Definition (P.4.1)**.** Let D and S be sets. A **function** $f : D \to S$ is a rule that assigns a *unique* element $f(x)$ in S to each element in D.

Usually also written as

$$\begin{aligned} f :& D \to S \\ & x \to f(x)\ . \end{aligned}$$

We call D the _______ of f $(D(f))$. If $x \in D$, then we call $f(x)$ the _____ of x. The set of all the images of all $x \in D$ is called the _____ of f $(R(f))$. In this calculus part, unless specified, functions will be $f : \mathbb{R} \to \mathbb{R}$.

Example. Some examples of real functions are

- $f(x) = x^2$, with $f(2) = 4, f(-2) = 4, f(3) = 9$. Its domain is $\mathbb{R}$ and the range is $\mathbb{R}_{\geq 0}$.
- $f(x) = |x|$, with $f(-3) = 3, f(2) = 2, f(0) = 0$. Its domain is $\mathbb{R}$ and the range is ___.
- $f(x) = \frac{1}{x}$, with $f(1) = 1, f(2) = \frac{1}{2}, f(-5) = -\frac{1}{5}$. Its domain is ______ and the range is $\mathbb{R}\backslash\{0\}$.

Do not confuse a function with its graph $G(f) = \{$____________________$\}$, which is the set of points in the domain with its images.

Definition (P.5.3)**.** If f, g are functions and x is in both their domains, then

- $(f + g)(x) = f(x) + g(x)$
- $(f - g)(x) = f(x) - g(x)$
- $(fg)(x) = f(x)g(x)$
- $(\frac{f}{g})(x) =$ ___, when ________.

Definition (P.5.4)**.** If $f : D \to S$ and $g : R \to D$ are functions, then $f \circ g : R \to S$ is the **composition** such that $(f \circ g)(x) =$ ______ . The domain of $f \circ g$ is made up of the values in the domain of g whose image is in the domain of f.

Example. Let $f(x) = \cos x$ and $g(x) = x^3$. Then $(f \circ g)(x) =$ ______.

An interesting kind of functions are piecewise defined functions, which take on different formulas on different parts of the domain.

Example.

$$f(x) = \begin{cases} x^2, & \text{if } \underline{\hspace{1.5em}} \\ -x, & \text{if } \underline{\hspace{1.5em}} \end{cases}$$

Polynomials

Definition (P.6.5)**.** A **polynomial** is a function $P : \mathbb{R} \to \mathbb{R}$ such that $P(x) = a_0 + a_1 x + \cdots + a_n x^n$, where $a_0, a_1, \cdots, a_n \in \mathbb{R}$ are the *coefficients* with $a_n \neq 0$ if $n > 0$. The number n is called the ______ of the polynomial.

Example. Some examples are

- $P(x) = 3$ is a polynomial of degree 0.
- $P(x) = 2 - x$ is a polynomial of degree 1.
- $P(x) = 3x^3 + 2x^2 - 1$ is a polynomial of degree 3.

Proposition. *Let P be a polynomial of degree n and Q a polynomial of degree m with $n \neq m$. Then:*

- *$P + Q$ has degree* __________.
- *PQ has degree* ______.

Definition (P.6.6)**.** Let P and Q be polynomials with $Q \neq 0$, then $R(x) = \frac{P(x)}{Q(x)}$ is a **rational function** whose domain is $\mathbb{R}$ except for the values x such that ________.

Example. The domain of $R(x) = \frac{3x^2+4}{2x-1}$ is $\mathbb{R} \backslash \{\frac{1}{2}\}$ since $2x - 1 = 0$ if and only if $x = 1/2$.

Definition (P.6)**.** A zero of a function f is a value $r \in \mathbb{R}$ such that ________.

A zero of a polynomial is also called *root*.

Example. A zero of $f(x) = x^3 - 3x^2 + 2$ is $x = 1$ since $f(1) = 1^3 - 3 \cdot 1^2 + 2 = 0$.

Theorem (P.6.1)**.** ***The Factor Theorem*** *The number r is a root of the polynomial P of degree not less than 1 if and only if* _____ *is a factor of $P(x)$.*

Example. From the previous example, $f(x) = x^3 - 3x^2 + 2 = (x - 1)(x^2 - 2x - 2)$.

Sometimes, it is useful to factor a polynomial, for instance, when you know a root and want to know the rest. This can come in handy in certain situations, for example, for computing eigenvalues of matrices (more on that later on in the course). Here are some simple factoring rules:

- $ax^2 + bx = x(ax + b)$.
- $x^2 - a^2 = (x - a)(x + a)$.
- $x^2 + 2ax + a^2 = (x + a)^2$.
- $x^2 - 2ax + a^2 = (x - a)^2$.
- $x^3 - a^3 = \underline{\hspace{4cm}}$.

Trigonometric Functions

There are many ways to define a trigonometric function, usually involving the relationship between the sides and angles of a rectangle triangle. Here we will give a more general definition so that we can define trigonometric functions for any real number. We will measure angles in *radians*, as the length of the arc that starts at the point $(1, 0)$ and follows the unit circle counterclockwise (in such units, π radians $= 180°$).

Definition (P.7.8)**.** For any real t, the **cosine** of t (abbreviated $\cos t$) and the **sine** of t (abbreviated $\sin t$) are the x- and y-coordinates of the point P_t on the unit circle (where t is the angle formed at the origin by the positive x semiaxis and the segment connecting the origin and the point P_t).

$$\cos t = \text{the } x\text{-coordinate of } P_t$$
$$\sin t = \text{the } y\text{-coordinate of } P_t.$$

Remark. Notice that we are computing the sine and cosine of an angle in the circle, up to 2π, however, taking a larger angle simply means doing one full turn to the circle and some more. That remaining angle is computed by getting the remainder of the division of the original angle by 2π. This means that trigonometric functions are *periodic* (and hence the wavy shape of their graphs), that is,

$$\cos(t + 2\pi) = \underline{\hspace{1cm}}, \quad \sin(t + 2\pi) = \underline{\hspace{1cm}}.$$

Example.

$$\cos 0 = 1, \quad \cos \frac{\pi}{2} = 0, \quad \cos \pi = -1, \quad \cos\left(-\frac{\pi}{2}\right) = \cos \frac{3\pi}{2} = 0$$
$$\sin 0 = 0, \quad \sin \frac{\pi}{2} = 1, \quad \sin \pi = 0, \quad \sin\left(-\frac{\pi}{2}\right) = \sin \frac{3\pi}{2} = -1.$$

Definition (P.7.9)**.** The **tangent** of an angle t is defined in terms of sine and cosine as

$$\tan t = \frac{\quad}{\underline{\hspace{1cm}}}.$$

There are some useful identities that may come in handy in some computations:

- The range of both $f(x) = \cos x$ and $g(x) = \sin x$ is $\underline{\hspace{1.5cm}}$, that is, $-1 \leq \cos x, \sin x \leq 1$, for all $x \in \mathbb{R}$.
- $\cos^2 t + \sin^2 t = 1$.
- Cosine is an even function; $\cos(-t) = \cos t$.
- Sine is an odd function; $\sin(-t) = \underline{\hspace{1.5cm}}$.

Limits to a Point

In this lecture we give *non-formal* definitions of limits.

Definition (1.2.1)**.** We say that a function f has limit L at a point $x = a$ if $f(x)$ gets as close as we want to L when we take x close enough to a and write it as

$$\lim_{x \to a} f(x) = L.$$

Example. Limits to some functions:

- $\lim_{x \to a} x^2 = a^2$
- $\lim_{x \to a} c = \_$

Exercise. Find the following limits:

- $\lim_{x \to -2} \dfrac{x^2 + x - 2}{x^2 + 5x + 6}$
- $\lim_{x \to a} \dfrac{\frac{1}{x} - \frac{1}{a}}{x - a}$
- $\lim_{x \to 4} \dfrac{\sqrt{x} - 2}{x^2 - 16}$

Definition (1.2.2)**.** If f is defined on (b, a) (resp. (a, b)), we define the **left** (resp. **right**) **limit** of f as $\lim_{x \to a^-} f(x)$ (resp. $\lim_{x \to a^+} f(x)$) if x gets close to a from the left side (resp. right).

Example. Consider the function

$$\operatorname{sgn} x = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$

Then $\lim_{x \to 0^-} \operatorname{sgn} x = \_\_\_$ and $\lim_{x \to 0^+} \operatorname{sgn} x = \_$.

Theorem (1.2.1)**.**

$$\lim_{x \to a} f(x) = L \iff \lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = L.$$

Exercise. Compute the lateral limits at $x = 2$ for $f(x) = \frac{|x-2|}{x^2+x-6}$. What about $\lim_{x \to 2} f(x)$?

Theorem (1.2.2)**.** *Let f, g be functions such that $\lim_{x \to a} f(x) = L$ and $\lim_{x \to a} g(x) = M$ and $k \in \mathbb{R}$, then*

- $\lim_{x \to a} (f(x) + g(x)) = L + M$
- $\lim_{x \to a} (f(x) - g(x)) = L - M$
- $\lim_{x \to a} (f(x) \cdot g(x)) = L \cdot M$

- $\lim\limits_{x\to a} kf(x) = kL$
- $\lim\limits_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{L}{M}$, *if* $M \neq 0$
- $\lim\limits_{x\to a} (f(x))^{\frac{m}{n}} = L^{\frac{m}{n}}$, *provided* $L > 0$ *if* n *is even and* m *is odd and* $L \neq 0$ *if* $m < 0$
- *If* $f(x) \leq g(x)$, __________.

Theorem (1.2.4)**.** ***The Squeeze Theorem*** *Let* f, g, h *functions such that* ________________ $\forall x \in (b, c)$ *and let* $a \in (b, c)$. *Assume* $\lim\limits_{x\to a} f(x) = \lim\limits_{x\to a} h(x) = L$. *Then* __________.

Example. Since $3 - x^2 \leq u(x) \leq 3 + x^2$ $\forall x \neq 0$, $\lim_{x\to 0} 3 - x^2 = \lim_{x\to 0} 3 + x^2 = 3 \implies \lim_{x\to 0} u(x) = 3$.

Limits at Infinity and Infinite Limits

Definition (1.3.3)**.** Let f be defined over ______, if $f(x)$ is as close to L as we want by taking x large enough, then we say ____________. Analogously, ______________.

Exercise. Calculate $\lim\limits_{x\to\pm\infty} f(x)$, where $f(x) = \frac{x}{\sqrt{x^2+1}}$.

Example. For limits toward infinity in rational functions it is important to look for the term of highest degree in the denominator:

- $\lim\limits_{x\to\pm\infty} \dfrac{2x^2 - x + 3}{3x^2 + 5} =$ ______________________________ $= \dfrac{2}{3}$
- $\lim\limits_{x\to\pm\infty} \dfrac{5x + 2}{2x^3 - 1} =$ ____________ $= 0$.

Example. Limits to points can also lead to infinity:

$$\lim_{x\to 0} \frac{1}{x^2} = \infty.$$

Example. Careful, recall that if lateral limits don't coincide, the limit does not exist:

$$\lim_{x\to 0^+} \frac{1}{x} = \infty, \quad \lim_{x\to 0^-} \frac{1}{x} = -\infty.$$

Example. There are also limits to infinity that lead to infinity

$$\lim_{x\to\pm\infty} \frac{x^4 + 3x}{x^2 - x + 5} = \lim_{x\to\pm\infty} \frac{x^2 + \frac{3}{x}}{1 - \frac{1}{x} + \frac{5}{x^2}} = \_\_.$$

Example. We have to be mindful of the signs on certain limits

$$\lim_{x\to-\infty} \frac{x^3 - 5x + 7}{-x^2 + 3x - 2} = \lim_{x\to-\infty} \frac{x - \frac{5}{x} + \frac{7}{x^2}}{-1 + \frac{3}{x} - \frac{2}{x^2}} = \_\_\_.$$